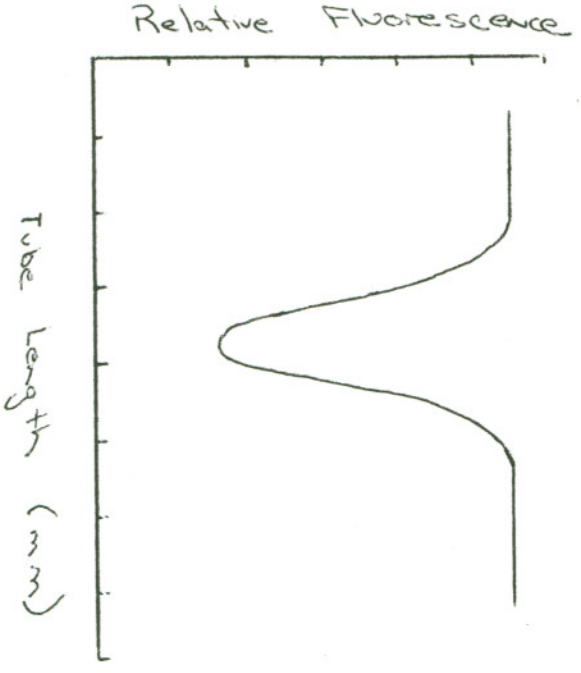
ANALYTICAL CUMULATIVE EXAMINATION
October 24, 1992

An unknown protein has been isolated and is to be characterized. Physical measurements show that it has a density of 1.33 g/cm^3, while size exclusion chromatography suggests that it has a molecular weight of 250,000 g/mol. An experiment is set up in which a specific mass of the protein is placed in the center of a 0.1 cm diameter tube which is very long compared to its diameter. The tube is filled with water containing a small amount of a fluorescent compound, sodium salicylate. The tube temperature is controlled at 15 ^{o}C . At this temperature, water has a viscosity of 0.0114 poises (g cm^{-1} s^{-1}). After 4 hours, a laser was scanned along the tube length and the fluorescence of the salicylate was monitored. At the exciting wavelength, the protein does not fluoresce. From this experiment, the data presented below were obtained.



a. Explain the data in terms of the shape of the curve and the rationale for the fluorescent signal observed.
b. Calculate the diffusion constant for the protein (cm^2/s). The tube is sufficiently narrow such that one can assume that the concentration gradient is zero across the tube diameter and only varies along its length.
c. What is the Stokes law radius of the molecule.
d. What is the radius calculated on the basis that the molecule is a sphere with the density noted above.
e. Discuss two reasons for any discrepancy noted between values calculated in b versus c.
f. Briefly describe Professor McGowns experimental technique. Suppose a fluorophor has been identified that is soluble in hydrophobic solvents and can be introduced into the interior of the above mentioned protein. Suggest an experiment, using lifetime resolved spectroscopy in which information could be obtained on the hydrophobic nature of the protein interior.

HELPFUL INFORMATION

$\sigma^2 = 2Dt$ $\qquad$ $C = \frac{1}{F} \frac{M}{\sigma_t \sqrt{2\pi}} e^{-\frac{1}{2}[(t_R - t)/\sigma_t]^2}$

$W_{0.5} = 2.354\,\sigma$ $\qquad$ $D = \frac{RT}{f}$ $\qquad$ $f = 6\pi\eta a$